Md. Ahasanul Adib

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scholar.google.com/citations ahsanadib.github.io/Md.AhasanulAdib.github.io

RESEARCH INTERESTS

- Power systems: distribution network modeling and AI-based grid optimization
- Integration of renewable energy into power grids
- Photonics and plasmonics: light-matter interaction and nanoparticle heating
- Deep learning for biomedical diagnosis

EDUCATION

BSc in Electrical and Electronic Engineering, *Islamic University of Technology* Jan 2020 – Jul 2024

- CGPA: 3.50 (Out of 4.00)

RESEARCH PUBLICATIONS

Deep Learning-Based Diagnostic Framework for Colorectal Cancer Using Histopathological Images, Jan 2025
Presented at IEEE ICAIIC 2025, Authors: Ashif Al Nayem Zeesan, Sadri Islam, Abrar Fahim, Md. Ahasanul Adib, Md. Touhidul Islam, Fardin Sabahat Khan, DOI: 10.1109/ICAIIC64266.2025.10920717

- Built a MobileNetV2 transfer-learning classifier on the LC25000 dataset. Applied data augmentation and preprocessing, including grayscale conversion, Gaussian filtering, thresholding, and pseudo-color mapping, to improve accuracy and generalization.
- Benchmarked against ResNet50, MA_ColonNET, and XGBoost. This showed how preprocessing and augmentation choices drive model accuracy.

CONFERENCE POSTER PRESENTATION

B.Sc. Thesis: Plasmonic Coupling and Thermal Effects on Photothermal Response of Randomly Distributed Nanoparticles, Nov 2024
International Conference on Centennial Celebration of Bose-Einstein Statistics: A Legacy of Dhaka. Authors: Hasnat Mohammad Nayeem, Abrar Fahim, Md. Ahasanul Adib

- Modeled light-driven heating in 1,000 randomly placed silver nanoparticles in water, using the discrete dipole approximation (DDA) for electromagnetic interactions and the thermal Green's function for temperature distribution.
- Found that plasmonic coupling strongly increases heating away from the resonance wavelength and reduces the temperature rise at resonance.

PROJECTS

License Plate Detection and Recognition System Sep 2022 – Dec 2022

- Built a MATLAB program that reads a car's license plate from an image. The program turns the image black and white, removes small unwanted spots, then crops and resizes the plate area to get clean letters and numbers.
- Used an ANFIS method to recognize each character and tested the system on about 30 real plate images for use in traffic monitoring and parking systems.

Simulation of a Regenerative Braking System in MATLAB Nov 2022 – Feb 2023

- Built a MATLAB model of a regenerative braking system that catches the energy normally lost during braking and sends it back to the battery, instead of letting it go to waste as heat.
- Used a DC shunt motor as a generator and a buck converter to control and store the recovered energy, choosing this motor over an induction motor for its better speed control and efficiency at partial loads.

Parkinson's Disease Prediction Using Machine Learning Oct 2023 – Dec 2023

- Worked with a dataset of 195 voice samples, each with 22 features like jitter, shimmer, and HNR. Cleaned and scaled the data, then trained four models (SVM, Logistic Regression, Gradient Boosting, and a Neural Network) and tested each one using 3-fold cross validation.
- The Neural Network gave the best result, reaching 89.7% accuracy and a 93.2% F1 score, better than the other three models on the same test data.

Economic Load Dispatch Using MATLAB Aug 2023 – Dec 2023

- Studied three common methods used in power systems to find the cheapest way to generate electricity (Lambda Iteration, Newton's Method, and Dynamic Programming) and built MATLAB programs to apply each one.
- Added generator limits and transmission losses to the model, then compared how fast and how accurate each method was, for both simple and more complex cost cases.

PROFESSIONAL EXPERIENCE

Anwer Khan Modern University, *Adjunct Faculty, Department of EEE* Oct 2026 – Present

- Teach undergraduate electrical engineering courses through lectures and laboratory sessions.
- Cover circuit analysis, AC and three-phase systems, and electrical machines, with a focus on problem-solving.
- Advise undergraduate students on academic progress and professional development.

Bangladesh Power Development Board, *Junior Engineer* Jul 2024 – Present

- Build distribution network models in ETAP from GIS data prepared in ArcGIS Pro
- Run load-flow and short-circuit studies, and write reports with recommendations on loss reduction and voltage regulation

Shikkhangon (Learning Academy), *Tutor & Mentor* Jun 2020 – Dec 2024

- Taught Physics and Mathematics and mentored students in problem-solving.

SKILLS

Simulation Software — MATLAB, ETAP, ArcGIS Pro, Proteus, PSpice, XG5000, PSIM, GeoGebra

Languages — Python (Pandas, Scikit-learn, PyQt5), C, LaTeX (Overleaf), HTML

Microsoft Office Tools — Word, PowerPoint, Excel

Interpersonal Skills — Critical Thinking, Organizational Skills, Mentoring, Leadership, Teamwork, Communication

STANDARDIZED TEST SCORE

IELTS

Reading: 8.5 Listening: 8 Speaking: **6** Writing: **6.5** Overall: **7.5**

TRAININGS

Bangladesh Power Development Board 2023

- Observed power generation & distribution operations at Ghorashal Power Station.

Institute of Nuclear Science and Technology 2023

- Explored nuclear reactor operation, machinery, and safety protocols.

Robi Axiata Limited 2023

- Engaged in telecom signal-transmission operations and infrastructure integration.

EXTRACURRICULARS

Head of Gaming (Operations), *Esonance 2023* 2023

- Led and managed gaming event operations (Chess, FIFA, Valorant) at the Esonance' 2023 fest, coordinating with participants from multiple universities.

Champion, *Annual Table Tennis, Doubles -2023* 2023

- Secured 1st place in Doubles at the Annual Table Tennis Tournament 2023, demonstrating strong teamwork and competitive spirit.

References

Dr. Rakibul Hasan Sagor, *Professor,* Islamic University of Technology
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